PROJECTION GEOMETRY

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OMFS 553

Mission: To advance the science and art of healthcare through education, service, research, personal wellness, and social accountability.

Overview

- Image sharpness and Resolution
- Image size Distortion
- Image shape Distortion
- Paralleling and Bisecting-Angle technique
- Object localization

- Conventional radiographs are 2D images produced with a stationary x-ray source.
    - Examples include intraoral, panoramic, cephalometric, and skull radiographs etc.
- High-quality 2D radiographs can allow reliable interpretation of 3D anatomy.
- Image quality reflects how accurately anatomy is represented.

Basic principles of projection geometry/shadow casting

- The focal spot should be as small as possible
- The source-receptor distance should be as long as possible
- The object-receptor distance should be as small as possible
- The receptor should be parallel to the long axis of the object
- The central beam should be perpendicular to the object and the receptor

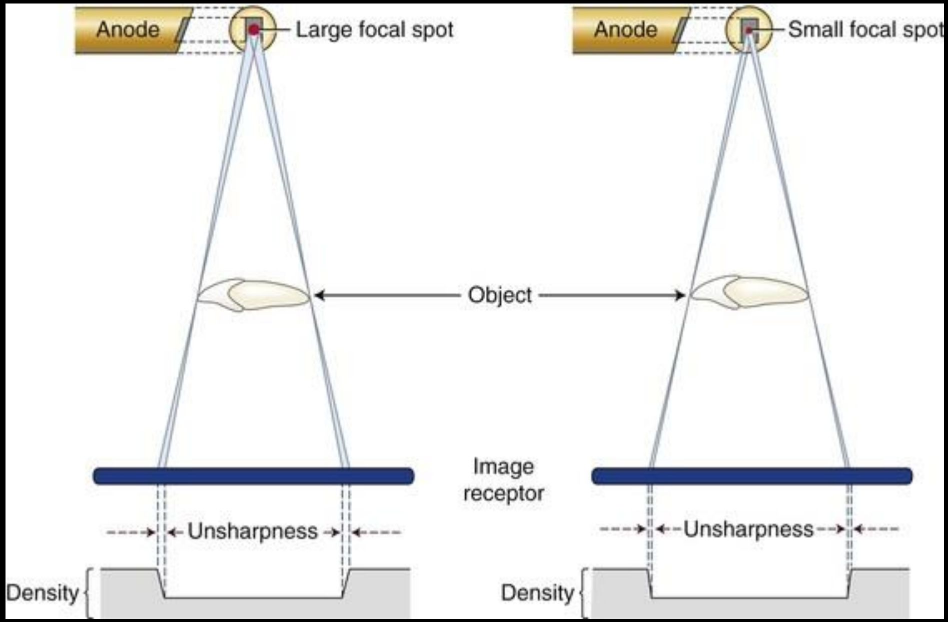
Image sharpness and resolution

- Image sharpness: How clearly boundaries between structures of different radiodensity are reproduced
- Spatial resolution: Ability to distinguish small objects that are close together
- Sharpness and resolution are related but not identical concepts
- Both are influenced by the same geometric factors
- Optimizing geometry improves diagnostic accuracy

Image sharpness and resolution

Penumbra (Geometric Unsharpness)

- X-ray photons originate from different points on the focal spot, creating zone of unsharpness (penumbra)
- Penumbra appears as a gradual density change at edges of enamel, dentin, and bone
- Larger focal spot → wider penumbra → less sharp image
- Smaller focal spot → narrower penumbra → sharper image
- Image sharpness is strongly dependent on focal spot size



Penumbra

Image sharpness and resolution

Maximizing Image Sharpness (1 of 3): Focal Spot Factors

- Use the smallest effective focal spot possible
  - Typical dental x-ray units: 0.4–0.7 mm
- Smaller focal spot → reduced penumbra → sharper image
- Angled x-ray tube target creates a smaller effective focal spot
- A target angle closer to the long axis of the electron beam:
  - Decreases effective focal spot size
  - Increases image sharpness
  - Improves heat dissipation but may shorten tube life

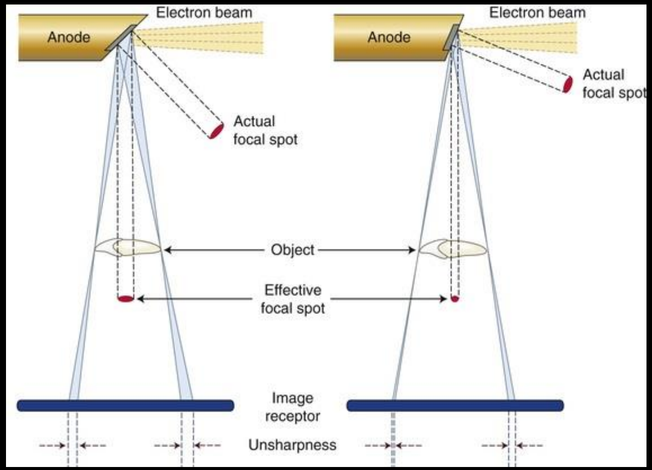


Image sharpness and resolution

Maximizing Image Sharpness (2 of 3): Source-Object Distance

- **Increase the distance between the x-ray source and the object**
- Achieved by using a long, open-ended aiming cylinder (long cone)
- Longer distance reduces beam divergence
- Reduced divergence → less geometric unsharpness
- Typical intraoral x-ray units: 8-16 inches from source to aiming ring

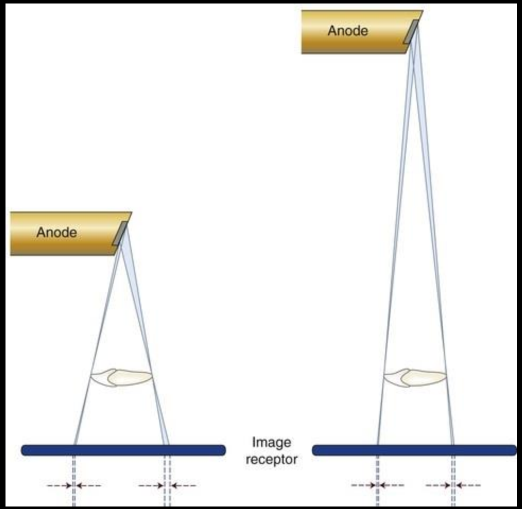


Image sharpness and resolution

Maximizing Image Sharpness (3 of 3): Object-Receptor Distance

- **Minimize the distance between the object and the image receptor**
- Shorter object-receptor distance reduces beam divergence at the object
- Reduced divergence → narrower penumbra
- Less penumbra → sharper image
- Achieved by proper receptor placement and positioning aids

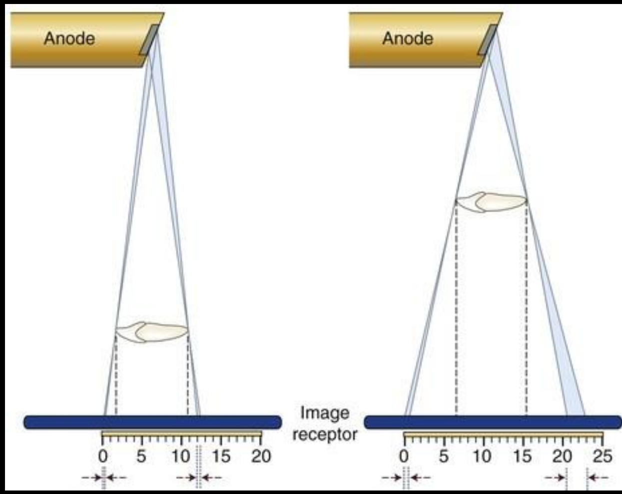


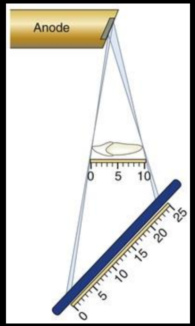
Image size distortion

Magnification

- Caused by divergent x-ray beam geometry
- Influenced by:
  - Source-object distance
  - Object-image receptor distance
- Greater object-receptor distance → greater magnification
- Reduced by using a long, open-ended aiming cylinder (long cone)

Image shape distortion

- Results from unequal magnification of different parts of the same object
- Occurs when the object, receptor, and x-ray beam are not properly aligned
- Common causes:
  - Object and image receptor not parallel
  - Central ray not perpendicular to the object and receptor
- Foreshortening: Excessive vertical angulation
- Elongation: Insufficient vertical angulation



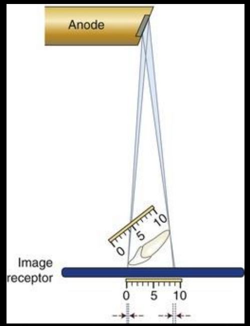
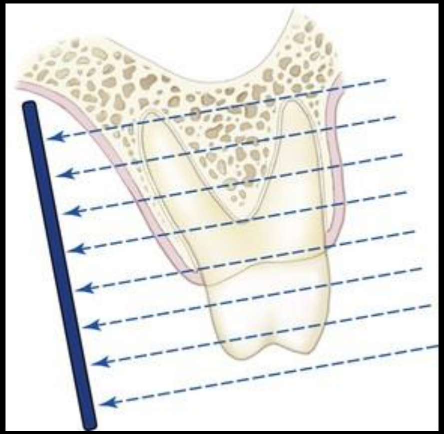


Image shape distortion

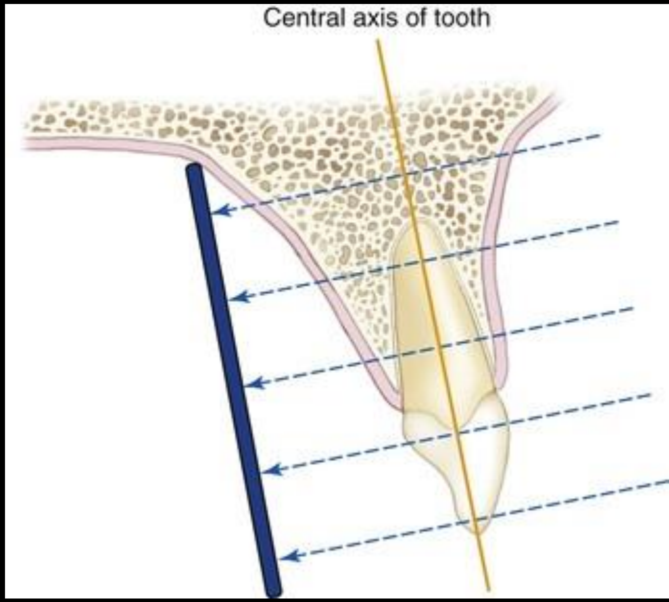


Central Ray Alignment

- Distortion occurs when the object and image receptor are parallel, but the central ray is not perpendicular to both
- Results in unequal magnification and distorted image shape
- Commonly seen with incorrect vertical angulation
- Most evident on maxillary molar radiographs

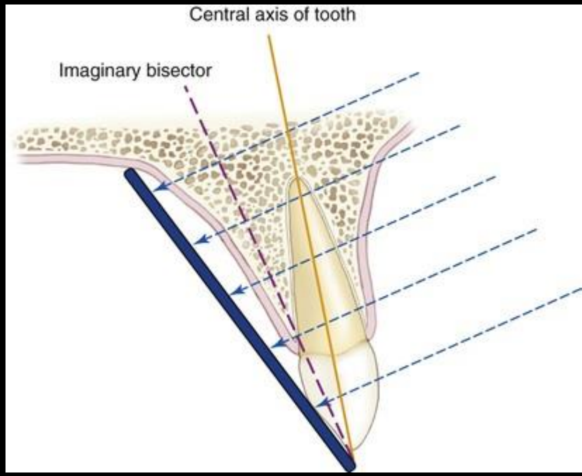
Paralleling Technique

- Preferred Intraoral Method
- Image receptor placed parallel to the long axis of the tooth
- Central x-ray beam directed perpendicular to both the tooth and the receptor
- Receptor positioned away from the teeth, toward the middle of the oral cavity
- Use a long, open-ended aiming cylinder (long cone)
- Receptor holders are required for accurate positioning



Bisecting-Angle Technique

- Image receptor placed as close to the tooth as possible
- An imaginary line bisects the angle between the tooth's long axis and the receptor
- Central x-ray beam directed perpendicular to this bisecting line
- Produces an image with approximately correct tooth length
- Results in more distortion than the paralleling technique
- Used less frequently today



OBJECT LOCALIZATION

- Based on plain (conventional) radiography
- Produces a 2-dimensional image of 3-dimensional structures
- All objects between the x-ray source and receptor are superimposed
- Used to determine the spatial relationship of structures, such as:
  - Impacted teeth
  - Foreign objects
  - Root canals
  - Fractures
- More useful in the mandible than the maxilla due to less superimposition

Approaches to decipher 3D relationships by radiography

- Right-angle technique (Miller's)
- Tube-shift technique / Buccal object rule/SLOB Rule (Clark's Rule)
- 3D modality, special studies

Right angle technique

- Requires at least two radiographs taken 90° to each other
- Images must be of the same area of interest
- Used to determine buccolingual position of structures

Intraoral approaches:

- Periapical view
- Topographic or cross-sectional occlusal view

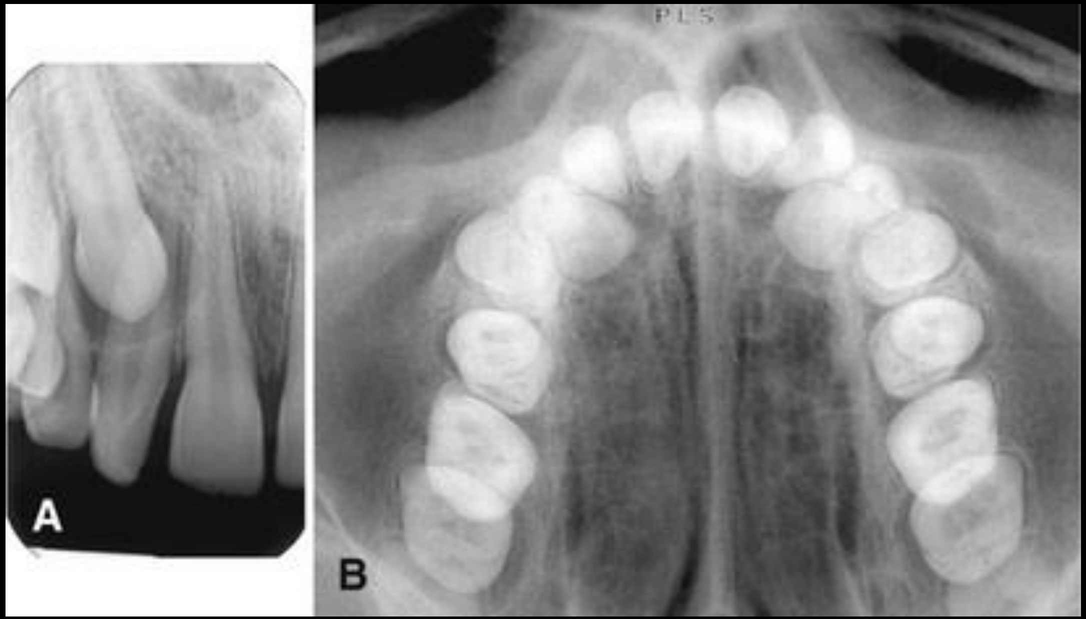
Extraoral approaches:

- Panoramic view
- Lateral oblique view
- Postero-anterior (PA) skull
- Lateral cephalometric
- Submentovertex (SMV) view

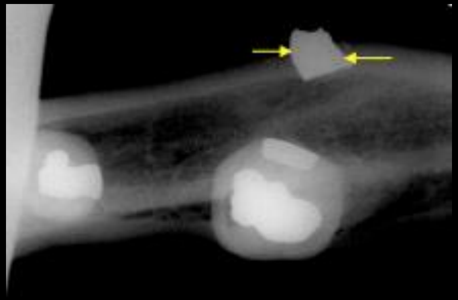
Right angle technique

Periapical Vs Cross-sectional view

- Periapical radiograph shows impacted canine lying apical to roots of lateral incisor and first premolar
- Maxillary cross-sectional occlusal view shows that the canine lies palatal to the roots of the lateral incisor and first premolar.

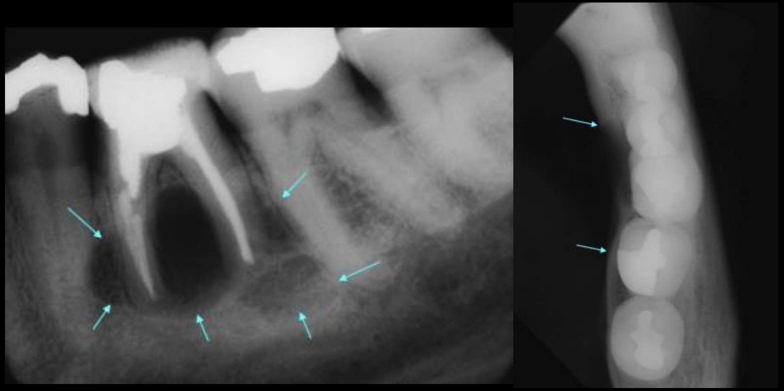


Right angle technique



Periapical Vs Cross-sectional view

- Periapical radiograph shows foreign object at mid root level of molar
- Mandibular cross-sectional view shows that the foreign object lies buccal to molar
- Periapical radiograph shows radiolucency apex of molar
- Mandibular cross-sectional view shows that the lesion lies lingual to molar

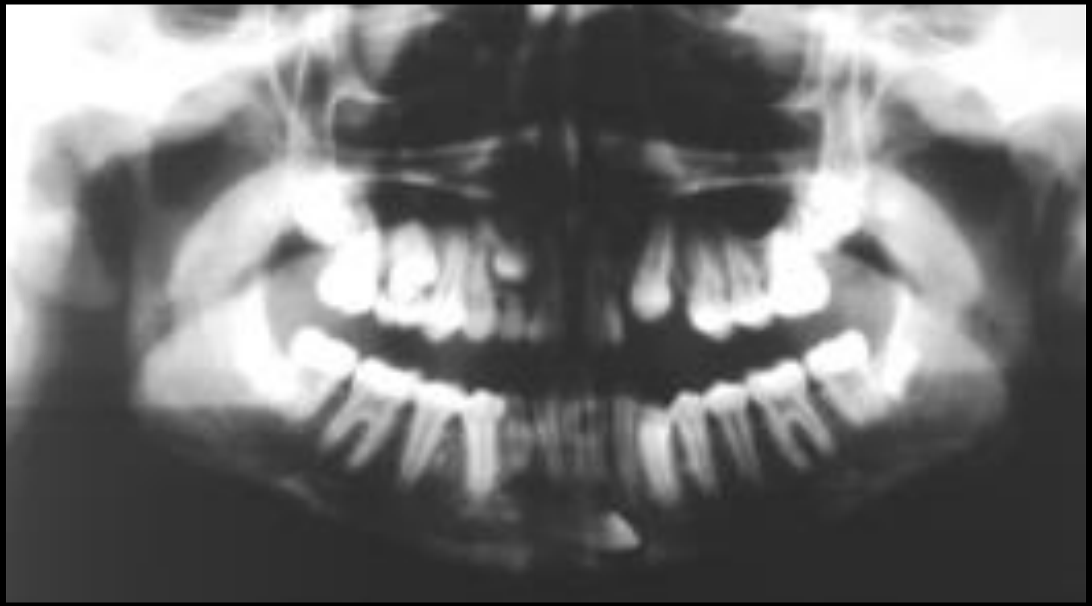


Right angle technique

Panoramic Vs Cross-sectional view

Right angle technique

Panoramic Vs Lateral skull view



Right angle technique

- PA Skull Vs Lateral Skull

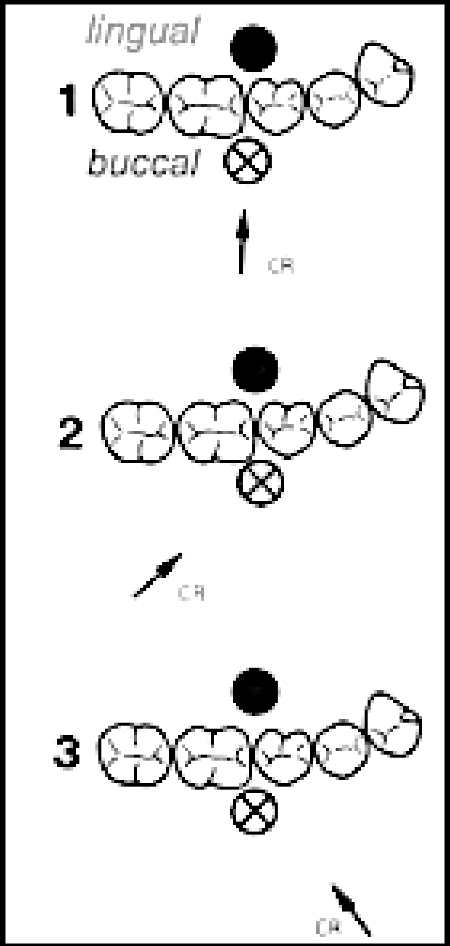
TUBE SHIFT TECHNIQUE

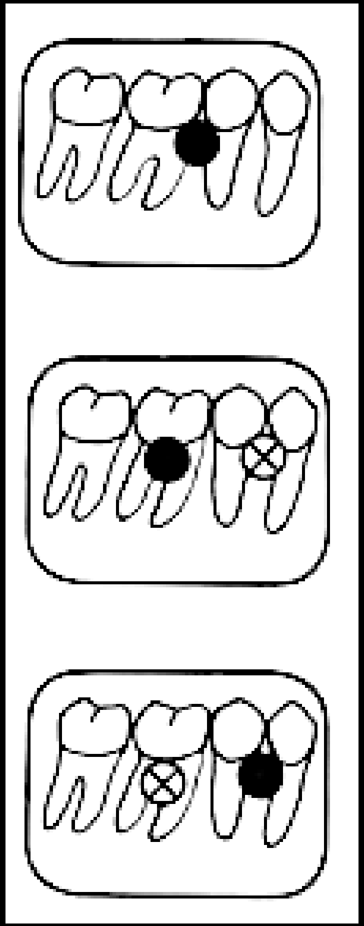
- Clark described this method in 1910
- Uses two radiographs with different beam angulations
- Image position changes when beam angulation changes
- Objects closer to the receptor move less; farther objects move more
- Used to determine buccal vs lingual position

SLOB

Same Lingual, Opposite Buccal

- Uses two radiographs with different beam angulations
- Lingual object moves in the same direction as the tube shift
- Buccal object moves in the opposite direction of the tube shift
- Used to determine buccal vs lingual position





- Usual paralleling technique
- The extra objects are superimposed on each other

- Tue shifted in the distal direction
- Notice how the lingual object moved in same direction
- While buccal object moved in mesial direction

SLOB